CCNA 200-301 V2.0

15 The Routing Table and How a Router Decides

Part III — IP Routing

EXAM TOPICS

3.1

LECTURER

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LEARNING OUTCOMES

What you will be able to do

- **Read** every column and code in `show ip route`.
- **Apply** the longest prefix match rule to determine which entry a given packet uses.
- **Explain** administrative distance and predict which source wins when two protocols offer the same prefix.
- **Distinguish** the three reasons a packet is dropped: no route, a route to null, and an unreachable next hop.

ASSUMED KNOWLEDGE

Before we start

Chapter 4 for prefixes and masks — this chapter is unreadable without fluent subnetting.
Chapter 10 for SVIs as gateways.

KEY TERMS

Vocabulary for this chapter

- routing table
- prefix
- next hop
- exit interface
- administrative distance
- metric
- longest prefix match
- directly connected
- local route
- default route
- gateway of last resort
- null0

Each is defined where it first matters — not here.

DEFINITION

Routing table

The list of destinations a router knows how to reach, and for each one, the **next hop** — the neighbouring device to hand the packet to — and the interface to send it out of.

A router does **not** know the whole path. It knows one step. Every router along the way makes the same one-step decision independently, and the packet arrives because each of them was right about its own next hop. That is worth sitting with, because it explains why a fault three hops away looks like a working router with traffic disappearing into it.

CONCEPT

One line, before twenty

Every entry in the table has the same shape. Read this one slowly and the full dump below stops being a wall of text:

```
O    10.1.0.0/16 [110/20] via 10.255.0.2, 00:14:22, GigabitEthernet0/1
```

- `O` — **how** the router learned it. `O` is OSPF, `S` static, `C` connected, `L` local. The legend at the top of the output lists them all.
- `10.1.0.0/16` — **what** it reaches: the destination network and its prefix length.
- `[110/20]` — **administrative distance** and **metric**. 110 is how much the router trusts the source; 20 is how good this particular path is. Section sec:ad explains why those are two separate numbers.
- `via 10.255.0.2` — the **next hop**.

CONCEPT

One line, before twenty (continued)

- `00:14:22` — how long the route has been known.
- `GigabitEthernet0/1` — the **exit interface**.

Connected and local routes have no distance, metric or next hop, because there is nothing between the router and them.

VERIFY

show ip route

R1# show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       O - OSPF, IA - OSPF inter area, N1 - OSPF NSSA external type 1
       E1 - OSPF external type 1, E2 - OSPF external type 2
       * - candidate default, U - per-user static route

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 203.0.113.1
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
O        10.1.0.0/16 [110/20] via 10.255.0.2, 00:14:22, GigabitEthernet0/1
C        10.255.0.0/30 is directly connected, GigabitEthernet0/1
L        10.255.0.1/32 is directly connected, GigabitEthernet0/1
S        10.9.9.0/24 [1/0] via 10.255.0.2
      192.168.10.0/24 is directly connected, Vlan10
C        192.168.10.0/24 is directly connected, Vlan10
L        192.168.10.1/32 is directly connected, Vlan10
```

CONCEPT

C and L are not the same thing

`C` is the *subnet* on a directly connected interface — the range the router can reach by ARP. `L` is a `/32` host route for the router's *own* address on that interface, so that traffic addressed to the router itself is delivered locally rather than forwarded.

You do not configure `L` routes; the router creates one for every address you give an interface. They appear in every table and are not a fault.

CONCEPT

The most specific route wins, regardless of anything else

When several entries could carry a packet, the router uses the one with the **longest prefix** — the most 1 bits in the mask. Administrative distance and metric never override this; they only choose between entries with the *same* prefix length.

WORKED EXAMPLE

Which entry carries the packet?

Using the table above, a packet for `10.9.9.20`:

- `0.0.0.0/0` — matches everything. Prefix length 0.
- `10.1.0.0/16` — does `10.9.9.20` fall inside? The /16 covers `10.1.0.0`–`10.1.255.255`. No.
- `10.9.9.0/24` — covers `10.9.9.0`–`10.9.9.255`. Yes. Prefix length 24.

Two entries match: the default and the /24. The /24 is longer, so the packet goes `via 10.255.0.2`. Note that the default route has a *better* administrative distance (1 versus 1 — equal here) and it still loses, because prefix length is decided first.

Now a packet for `172.20.5.1`: only `0.0.0.0/0` matches, so it takes the gateway of last resort.

DEFINITION

Administrative distance

Administrative distance ranks the trustworthiness of a route's *source*. When two sources offer the same prefix at the same length, the lower distance is installed and the other is discarded — it does not even appear in the table.

CONCEPT

Why a static route beats OSPF, and when that bites

A static route at distance 1 will always displace an OSPF route at 110 for the same prefix. That is usually what you want — until the static route's path fails. OSPF would have re-routed; the static route sits there pointing at a dead path, because a static route has no way of knowing anything is wrong beyond the directly connected interface.

The deliberate use of this is the **floating static route**: a backup static configured with a distance *higher* than the dynamic protocol, so it stays out of the table until the dynamic route disappears. That is Chapter 16.

WATCH OUT

A route in the table is not proof of reachability

The presence of an entry means the router knows where it *would* send the packet. It says nothing about whether the next hop is alive, whether the far end has a route back, or whether an access list drops it en route.

Confirm with an extended ping sourced from the relevant interface (Chapter 6) — that tests the return path too, which a route entry never does.

VERIFY

show ip route 10.9.9.20

R1# show ip route 10.9.9.20

```
Routing entry for 10.9.9.0/24
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
  * 10.255.0.2
      Route metric is 0, traffic share count is 1
```

Three ways a packet dies

- **Comparing metrics across protocols.** An OSPF cost of 20 and a RIP hop count of 2 are not comparable. Administrative distance decides first, and only then does metric compare *within* one protocol.
- **Thinking a better distance beats a longer prefix.** It does not. Prefix length is decided first, always.
- **Missing the gateway of last resort line.** If it says `not set`, there is no default and anything unmatched is dropped.
- **Reading `L` entries as duplicates.** They are the router's own addresses and belong there.

Three ways a packet dies (continued)

- **Assuming symmetry.** Traffic reaching a destination does not mean the reply can return. Most one-way faults are a missing reverse route.

The route is there and the ping still fails.

Users on `192.168.10.0/24` cannot reach `10.9.9.0/24`. `R1` has the route:

A ping from `R1` to `10.9.9.20` succeeds. A ping from a user's PC fails.

The route is there and the ping still fails.

What would you check first — and what do you expect to see?

show ip route 10.9.9.0

R1# show ip route 10.9.9.0

```
Routing entry for 10.9.9.0/24
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
  * 10.255.0.2
```

ping 10.9.9.20 source Vlan10

R1# ping 10.9.9.20 source Vlan10

```
Sending 5, 100-byte ICMP Echos to 10.9.9.20, timeout is 2 seconds:
Packet sent with a source address of 192.168.10.1
.....
Success rate is 0 percent (0/5)
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. The difference between the two pings is the source address. `R1`'s own ping is sourced from `10.255.0.1`, which the far end knows about because it is the link between them. The PC's traffic is sourced from `192.168.10.x`, and the far router has no route back to that subnet. The forward path is fine; the return path does not exist.

Confirm it without leaving `R1`, using an extended ping that borrows the user's source address:

Same destination, different source, opposite result — which is proof, not inference.

And the lesson

Fix. Add a route on the far router back to `192.168.10.0/24`. Then re-run the same extended ping and expect five exclamation marks.

Read the table, then predict the path

Topology. Three routers in a line, a LAN on each end, and a second path between the outer two.

1. Address everything and confirm the connected routes appear. Identify every `C` and `L` entry and explain each `L`.
2. Add static routes so the two LANs can reach each other. Record the distance and metric shown.
3. Add a default route on one router. Note the `S*` and the *gateway of last resort* line.
4. Write down, before testing, which entry a packet to each of four addresses would use. Verify each with `show ip route <address>`.
5. Add a second, more specific route for one destination and confirm it wins over the less specific one, regardless of distance.

Read the table, then predict the path (continued)

1. Remove the reverse route on the far router and reproduce the one-way fault above using `ping ... source`.
2. **Extension.** Add a route pointing to `Null0` for an unused prefix and observe how traffic to it fails compared with having no route at all.

CHECKPOINT 1 of 3

A table contains `10.0.0.0/8` via OSPF and `10.5.0.0/16` via a static route. Which carries a packet to `10.5.1.1`, and why?

Discuss · then advance

ANSWER

Checkpoint 1 of 3

A table contains `10.0.0.0/8` via OSPF and `10.5.0.0/16` via a static route. Which carries a packet to `10.5.1.1`, and why?

> The static `10.5.0.0/16`, because it is the longer prefix match. Administrative distance never enters into it — longest prefix match is applied first and decides between *entries*.

Both OSPF and a static route offer `192.168.50.0/24`. Which is installed, and what happens to the other?

Discuss · then advance

ANSWER

Checkpoint 2 of 3

Both OSPF and a static route offer `192.168.50.0/24`. Which is installed, and what happens to the other?

> The static route, distance 1 against OSPF's 110. The OSPF route is not installed in the routing table but remains in the OSPF database, so it will appear if the static route is removed or its next hop becomes unreachable.

CHECKPOINT 3 of 3

A route exists but pings fail. Name three causes and the single command that distinguishes the most common one.

Discuss · then advance

ANSWER

Checkpoint 3 of 3

A route exists but pings fail. Name three causes and the single command that distinguishes the most common one.

> A layer 2 problem on the path; an ACL blocking the traffic; or no return route at the far end. `traceroute` distinguishes the most common by showing where the packets stop, and whether they get there and cannot get back.

What to take away

- One table, many sources. `C` connected, `L` the router's own address, `S` static, `O` OSPF.
- `[distance/metric]`. Distance ranks the source; metric ranks paths within one protocol and is never compared across protocols.
- Longest prefix match decides first. Distance only breaks ties between equal-length prefixes.
- Connected 0, static 1, OSPF 110. A static route displaces OSPF and will not fail over unless you make it float.
- A route in the table proves intent, not reachability. Test the return path with an extended ping.

Where we got to

- One table, many sources. `C` connected, `L` the router's own address, `S` static, `O` OSPF.
- `[distance/metric]`. Distance ranks the source; metric ranks paths within one protocol and is never compared across protocols.
- Longest prefix match decides first. Distance only breaks ties between equal-length prefixes.
- Connected 0, static 1, OSPF 110. A static route displaces OSPF and will not fail over unless you make it float.

NEXT

Chapter 16 — Static Routing, IPv4 and IPv6